Recombination comes into play for pulse durations longer than the thermalization time of CB electrons, and thermal ionization plays a role when recombination and collisional dissipation have lasted long enough to heat the focus to temperatures ≥ 3000 K [159]. It can be seen from Fig. 1 that recombination and thermal ionization counteract each other – while recombination inhibits the ionization process for long laser pulses, it is enhanced by thermal ionization. A consideration of their interplay reveals interesting features that have been overlooked in the past.

While consideration of details is required to explore and quantify the shades of optical breakdown dynamics, the simple form of Eqs. (1) – (4) helps in obtaining a qualitative understanding of the great picture in the entire range of pulse durations from fs to ns. It enables to anticipate and characterize general scenarios for the irradiance and energy dependence of the breakdown behavior in the (τ_L, λ_L) parameter space. The next section describes three qualitatively different scenarios covering the range from low-density to high-density plasma formation. Two scenarios are well known from previous research, and a third one is predicted by analyzing the interplay of nonlinear energy deposition, recombination, and thermalization pathways. The detailed features of these scenarios and their borders in the (τ_L, λ_L) space will then be explored experimentally.

2.2. Breakdown scenarios from precisely tunable energy deposition to "big bang"

Scenario 1: Continuous tunability such as observed in fs breakdown. At ultrashort pulse durations, the breakdown dynamics is governed solely by strong-field and avalanche ionization. Recombination and thermalization can be neglected since they occur on a ps time scale after the laser pulse. Strong-field-created seed electrons are abundant but the value of ρ_c at the end of the laser pulse depends mainly on the avalanche ionization rate, as discussed in the previous section. Since the avalanche ionization rate is limited by time constraints, ρ_c increases only gradually with growing irradiance [Eq. (2)]. Therefore, the onset of breakdown is smooth, and the amount of deposited energy – and the bubble size - can be continuously tuned by varying E_L.

Scenario 2: "Big bang" such as observed in IR ns breakdown. With increasing laser pulse duration, more and more doubling sequences of the ionization avalanche can occur during a pulse. Therefore, the irradiance threshold for bubble formation could largely decrease for long pulses if seed electrons for avalanche ionization were readily available. However, seed electron generation by multiphoton ionization requires a minimum irradiance. This hurdle is particularly large for long wavelengths, where high-order multiphoton processes are required for crossing the bandgap. Thus, for IR ns pulses seed electron generation by multiphoton ionization is the critical hurdle for the occurrence of breakdown [36].

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Once this hurdle is overcome, the ionization avalanche proceeds very fast. In conjunction with thermal ionization, it can largely overshoot the threshold condition and produce bright plasma luminescence and a large bubble already at the onset of breakdown. In this "big bang" scenario, tunability of energy deposition is restricted to the high-density plasma regime.

Scenario 3: Stepwise tunability in ps and ns breakdown. At short wavelengths, multiphoton ionization always provides sufficient seed electrons for AI, even at nanosecond pulse durations. However, although the ionization avalanche has an easy start, it is considerably slower than in IR breakdown because the avalanche ionization rate is smaller for shorter wavelengths as it is approximately proportional to λ^2 [37]. The slow ionization avalanche will thus be inhibited by recombination because the recombination rate is proportional to ρ_c^2 [Eq. (3)], while the avalanche ionization rate exhibits a linear dependence on ρ_c [Eq. (2)]. In the lower irradiance range near threshold, the growth of the free electron density is limited and a dynamic equilibrium between avalanche ionization and recombination evolves in which laser energy is continuously deposited into the electronic system and thermalized simultaneously. The focal temperature reached at the end of the laser pulse depends on the equilibrium level of ρ_c, which is determined by the focal irradiance. The above description implies that ns pulses at UV wavelengths and possibly also at short visible (VIS) wavelengths fulfill essential prerequisites for precisely tunable energy deposition and the creation of nano- and micro effects – a feature, which traditionally has been ascribed to ultrashort laser pulses and scenario 1.

With increasing irradiance (or E_L, respectively), the ionization avalanche produces an ever higher free electron density before it is inhibited by recombination. Thus, the focal temperature reached at the end of the pulse increases gradually as long as recombination can hold the ionization process. Beyond a critical temperature, the breakdown dynamics will change dramatically due to the exponential dependence of thermal ionization on temperature [Eq. (4)]. Together with avalanche ionization, thermal ionization now overcomes the inhibiting action of recombination, and ρ_c can suddenly shoot up to very high ionization levels. Consequently, brightly luminescent plasma with high energy density is formed. The existence of a second breakdown threshold at the upper end of the nano/micro regime is a key difference to scenario 1 with continuous tunability. Thus, scenario 3 is an intermediate between the other better-known scenarios, sharing features of scenario 1 at low irradiance and of scenario 2 at high irradiance. The abrupt transition between the tunable low- and high-density plasma regimes goes along with a sudden, stepwise increase of the vaporized liquid volume, plasma pressure and bubble size.

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The tunable energy deposition and creation of nanoeffects in the low-energy part of scenario 3 opens interesting perspectives for the use of compact diode-pumped UV solid state lasers in applications for which to date fs lasers are employed. Some evidence for the possibility of creating nano- and microeffects with tightly focused UV-A ns lasers has been reported previously [192,193]. However, this possibility has not yet been systematically explored because of the widespread view that only fs pulses allow for reproducible ("deterministic") energy deposition whereas ns breakdown exhibits "stochastic" character and is associated with vigorous laser effects [22,127,129,194,195]. In the present paper, we investigate the prerequisites for deterministic breakdown behavior and explore, in which wavelength range nanoeffects by non-luminescent ns plasmas can be produced.

2.3. Deterministic vs. stochastic breakdown behavior

For ultrashort laser pulses, where seed electrons are abundant, breakdown dynamics and bubble threshold show little dependence on impurities as long as pulse duration and energy remain stable. In these regions, the breakdown dynamics is, therefore, "inherently" deterministic. By contrast, when MPI initiation is the critical hurdle for the occurrence of breakdown, the breakdown dynamics depends more sensitively on small variations of the initial conditions. Here, a deterministic behavior can be expected only for highly reproducible laser pulses with smooth temporal profile and good beam quality, and for largely impurity-free media. While the temporal profile of mode-locked laser pulses is fairly stable, many solid state ns lasers are run in longitudinal multimode operation in which the pulse shape exhibits intensity spikes arising from statistical interference of the longitudinal modes that affect the multiphoton excitation rate [Eq. (1)]. Pulse-to-pulse fluctuations of the spiking behavior with intensity peaks of varying height at different times during the pulse introduce strong fluctuations in seed electron generation that result in a stochastic breakdown behavior.

Impurities can also influence the seed electron generation - either by localized heating of tiny particles followed by thermionic electron emission or by providing intermediate energy levels in the water band gap that facilitate multiphoton excitation. Contaminations of low concentration in a nominally 'pure' medium differ from biomolecules in aqueous biological media that provide numerous centers of reduced excitation energy at high concentration [196-198]. Both thermionic emission of seed electrons and/or reduced excitation energy for MPI at impurities in biological tissues can lower I_{th} in IR ps and ns breakdown [109,199]. By contrast, in UV breakdown, seed electrons are readily available, and significant changes of I_{th} can here only arise when the impurity concentration is so high that it significantly widens the MPI-channel [37], or when the seed electrons become available at fluctuating

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